CANE WATER AUDIT

Use in the Distillery and Site

Guidance for recording daily water use, identifying losses and improving reuse

1. Purpose and scope

This section of the Cane Water Audit records water used in the distillery and supporting site services. The four daily fields are: (1) Process and mash dilution, (2) Cooling tower make-up, (3) DM plant and boiler, and (4) Domestic and green belt. Enter the actual measured quantity in litres per day (L/day). Use a consistent 24-hour reporting period and record the meter readings, operating hours and production for that day.

Important: these fields are intended to show water entering or being added to each system. Do not count the same water twice. For example, water recovered from condensate and reused in mash preparation should be recorded as reused/recovered water in the plant's water balance, not treated as new freshwater withdrawal. Follow the definitions used by the audit tool and keep separate totals for freshwater, recovered water and wastewater wherever possible.

2. Explanation of the four fields

A. Process and mash dilution — L/day

What it covers: Water used directly in preparing or diluting the distillery process stream. Depending on the plant configuration, this may include dilution water for molasses/syrup or other feedstock, mash/wash preparation, adjustment of fermentable-sugar concentration, and process water added at designated mixing points. Include only water actually added to the process; do not automatically include cleaning water or cooling-tower water here.

How to measure: Read the dedicated flow meter on the process/mash dilution line. If there is no meter, estimate from a calibrated tank: volume added per batch × number of batches per day. Record the source (freshwater, condensate, treated/recovered water) separately.

Audit checks and improvement opportunities: Compare water use with feedstock processed and fermentation requirements. Check dilution ratios and process specifications with the plant process team; avoid reducing water in a way that affects mixing, sugar concentration, fermentation or product quality. Look for leaking valves, overflow, unmetered hoses and opportunities to reuse suitable condensate or treated water after quality approval.

Example calculation: If a calibrated tank supplies 8 m^3 per batch for 12 batches/day, use = 8 × 12 = 96 m^3/day = 96,000 L/day. This is an arithmetic example only, not a recommended distillery consumption rate.

B. Cooling tower make-up — L/day

What it covers: Fresh or recovered water added to the cooling-tower circuit to replace water lost through evaporation, drift, blowdown and leakage. Record make-up water entering the cooling tower system, not the circulating water repeatedly moving through the cooling circuit.

How to measure: Use a make-up water flow meter installed on the cooling-tower inlet. If multiple towers serve different units, measure each tower and sum the daily make-up. Record blowdown, conductivity/TDS, cycles of concentration (COC), operating hours and any overflow where available.

Audit checks and improvement opportunities: Check float valves, basin overflow, drift eliminators, leaks and blowdown control. Verify water-treatment limits before adjusting COC. Consider suitable treated wastewater or condensate only after checking chemistry and compatibility with the cooling-water treatment programme. Do not reduce blowdown below safe treatment limits.

Example calculation: A meter reading increases from 125.0 m³ to 171.5 m³ over the reporting day: make-up = 46.5 m³/day = 46,500 L/day. Use the meter difference, not the tower's recirculation-pump flow.

C. DM plant and boiler — L/day

What it covers: Water used by the demineralisation (DM) plant to produce demineralised water for the boiler and related approved users, together with boiler make-up water as captured by the site's measurement arrangement. This field must be defined consistently: if the audit form asks for the DM-plant feed, record feed to the DM plant; if it asks for boiler/DM water consumption combined, record the agreed combined quantity and document the boundary.

How to measure: Ideally record separate meters for raw-water feed to the DM plant, DM product water to the boiler/DM tank, and boiler make-up. Also record regeneration/rinse water and reject streams if metered. Do not add both DM product water and boiler make-up together if they represent the same water at two points in the same flow path.

Audit checks and improvement opportunities: Review DM recovery, regeneration frequency, reject-water quantity and quality, condensate return, steam leaks, boiler blowdown and condensate contamination. Reuse reject water only for a suitable non-critical purpose after checking quality and operational requirements. Boiler feedwater quality must meet the boiler supplier's and plant's specifications.

Example calculation: If 72 m³/day enters the DM plant and 54 m³/day of DM product is generated, the apparent product recovery is 54 ÷ 72 × 100 = 75%. Record the 72 m³/day feed and 54 m³/day product separately if the audit supports sub-meters; do not report their sum as 126 m³/day of unique water use.

D. Domestic and green belt — L/day

What it covers: Water for toilets, washrooms, handwashing, canteen and other staff/visitor domestic uses, plus irrigation of lawns, gardens, trees and other green-belt areas. Where possible, keep domestic and irrigation sub-meters separate even if the audit interface combines them into one field.

How to measure: Read the domestic water meter and irrigation meter at the same start and end times. If there is only one combined meter, record the combined total and state that the split is unavailable. For tanker supply, record tanker volume from invoices or calibrated capacity and count only the water delivered during the reporting day.

Audit checks and improvement opportunities: Check leaking taps and flush valves, tank overflows and hose use. Consider low-flow fixtures and rainwater harvesting. Schedule irrigation during cooler hours and use drip irrigation or mulch where suitable. Use treated wastewater for landscaping only when its quality, local requirements and site safety arrangements permit it.

Example calculation: Domestic use of 14 m³/day plus irrigation of 9 m³/day gives a combined value of 23 m³/day = 23,000 L/day. These figures are illustrative only.

3. Daily recording sheet

Audit field	Meter / source	Start reading	End reading	Daily quantity (L/day)
Process and mash dilution	Dedicated process meter / batch tank			
Cooling tower make-up	Cooling-tower make-up meter			
DM plant and boiler	DM feed/product/boiler make-up meters			
Domestic and green belt	Domestic and irrigation meters / tanker log			

For a meter measuring cubic metres (m³): daily quantity in L/day = (end reading − start reading) × 1,000. For a meter measuring kilolitres (kL), 1 kL = 1,000 L. For a batch tank, daily quantity = measured volume per batch × number of batches.

4. Illustrative daily summary (not a benchmark)

Field	Illustrative quantity (L/day)	Illustrative quantity (m³/day)

Process and mash dilution	96,000	96.0
Cooling tower make-up	46,500	46.5
DM plant and boiler	To be measured	To be measured
Domestic and green belt	23,000	23.0

The numbers above are simple examples to demonstrate recording and conversion only. They are not recommended consumption values for any distillery. Actual water demand varies with plant capacity, feedstock, product mix, cooling design, boiler pressure, condensate return, operating hours, weather and water-treatment configuration.

5. Good practice for reliable audit data

- Use the same reporting period and units for all readings; note shutdowns, start-ups and abnormal events.
- Record plant throughput alongside water data (for example, tonnes of feedstock processed and litres of ethanol produced) so that water intensity can be calculated.
- Keep freshwater intake, recovered/reused water, internal recirculation and wastewater discharge as separate categories.
- Install or verify meters at system boundaries; calibrate meters periodically and retain dated readings or photographs.
- Investigate sudden changes, unexplained differences and meter readings that do not agree with the site water balance.
- Do not assume that all water streams are interchangeable. Confirm water quality and process, boiler, cooling or environmental requirements before reuse.

6. Suggested audit indicators

Indicator	Calculation	Purpose
Daily water use	Sum of defined, non-duplicated daily water inputs	Track day-to-day consumption
Water intensity	Water quantity (m^3/day) ÷ production per day	Compare performance across operating days; state the chosen denominator
DM product recovery	DM product volume ÷ DM feed volume × 100	Monitor DM plant performance
Reuse share	Recovered/reused water ÷ total water supplied to defined uses × 100	Track substitution of freshwater, using a clearly stated boundary

Boundary note: Before using these four entries in a total, confirm how the Cane Water Audit tool defines each field. In particular, DM plant feed and boiler make-up can describe the same water at different points; adding both without a defined boundary can double-count it. The same principle applies when recovered water is reused in process or cooling systems.